Constructor University

Datathon 2026

Causation, Correlation & Causal Inference

April 17–19, 2026 • Constructor University, Bremen

Initial / Draft Program Schedule

Event Overview

The Constructor University Datathon 2026 brings together students and researchers to tackle real-world problems rooted in causal inference, correlation vs. causation theory, directed acyclic graphs (DAGs), and probability. Participants will work in teams over three days to explore, model, and present data-driven solutions grounded in causal reasoning.

Industry partner Insytes GmbH will provide the challenge problem and support throughout the event. Academic mentors and judges are drawn from Constructor University faculty with expertise in causality and probabilistic modelling.

What We Want from External Collaborators

This section outlines the expected contributions from each external partner and collaborator.

Insytes GmbH (Industry Partner)

- Define and present the challenge problem (data background, real-world context, constraints, deliverables)
- Provide the dataset(s) and any necessary domain context
- Offer mentoring and technical support to teams throughout the datathon
- Participate in the Panel Discussion on Day 1
- Serve as a judge during final team presentations on Day 3

Professors: Keivan Mallahi-Karai & Ivan Ovysannikov

- Pre-event: Review and validate the challenge problem statement to ensure academic rigour and alignment with causal inference / DAG / probability themes
- Day 2: Deliver the Academic Talk on methods in causal inference and probabilistic modelling
- Days 1–3: Available as mentors and advisors during team work sessions
- Day 3: Serve as judges during final team presentations

⚠ *Role assignment (who gives the talk, who leads pre-event review) to be confirmed between Keivan and Ivan.*

Omar Elshinawy

- Day 1: Deliver the Keynote Talk on Artificial Intelligence (Title: Automated Theorem Proving Using Causal ML)
- Day 1: Participate in the Panel Discussion on AI and industry–academia collaboration

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Day 1 — April 17: Opening, Talks & Challenge Launch

Day 1 focuses on orientation, context-setting, and launching the challenge. The challenge introduction and technical briefing run back to back so teams have a complete picture before the keynote, academic talk and panel. Teams begin working in the afternoon.

14:00–14:30	**Registration & Participant Check-in** • Team arrival and registration • Distribution of event materials (badges, schedule, team identifiers) ⚠ *Maha: Who is preparing the badges and printing the schedule?* ⚠ *Maha: Who will be the host / MC?*
14:30–14:45	**Opening Remarks & Event Introduction** *Speaker: Organiser / Datathon Lead* • Welcome message • Overview of the Datathon • Explanation of competition format • Introduction of partners and sponsors
14:45–15:15	**Keynote Talk — Automated Theorem Prooving** *Speaker: Omar Elshinawy (Automated Theorem Proving Using Causal ML)*
15:15–15:35	**Academic Talk —Introduction to Causal AI** *Speakers: Varun Patel, one of the CEOs of Insytes (Online)* • Causal inference frameworks: do-calculus, interventions, and counterfactuals • Directed Acyclic Graphs (DAGs): construction, d-separation, and identifiability • Correlation vs. causation: common pitfalls and how to avoid them • Probabilistic modelling techniques relevant to the challenge ⚠ *Maha: title and abstract TBC*
15:35–16:05	**Panel Discussion — From Proof to Product: How Mathematical Reasoning Shapes Industry** • Panelists: Omar Elshinawy, Representative from Insytes GmbH, Karim, Prof Keivan, Ahmed Maghri • Topics: AI in industry, data science careers, academic–industry collaboration • Causal thinking and its role in AI applications ⚠ *Maha: confirm moderator* ⚠ *Maha: confirm whether any faculty will join the panel*
16:05–16:35	**Coffee Break & Networking**
16:35–16:55	**Introduction of the Challenge Problem** *Speaker: Representative from Insytes GmbH: Karim Or Ahmed* • Background of the data problem and its causal inference context • Real-world application of causation and correlation analysis

	• Data sources, variables, and constraints • Expected deliverables from teams

16:55–17:15	**Technical Briefing for Teams** *Speaker: Ahmed* • Rules of the Datathon • Submission format and platform • Evaluation criteria • Available datasets and tools ⚠ *Maha: confirm whether technical briefing is delivered by the host or the company*

17:15–19:00	**Team Work Session 1 — Problem Exploration** • Read and understand the challenge problem and causal context • Explore and clean the dataset (handle missing values, noise, outliers) • Identify key variables and hypothesise causal/correlational relationships • Sketch initial DAG structures and probabilistic assumptions • Brainstorm solution approaches ⚠ *Mentors: to be confirmed (Insytes GmbH and/or faculty)* ⚠ *Technical assistance: to be confirmed (organisers and/or company)*

Day 2 — April 18: Team Work & Support

Day 2 is primarily a free working day for teams, focused on independent development of solutions

09:00–12:00	**Team Work Session 2 — Model Development** • Formalise causal hypotheses using DAG structures • Develop and implement causal / probabilistic models • Iterative testing, validation, and refinement of models • Begin preparing visualisations

11:00–13:00	**In-person Q&A & Support Session (Optional)** • Students can drop in to ask questions or request support • Informal discussion with available mentors **Location: IRC Seminar Room 3**

12:00–13:00	**Lunch Break (self-arranged)**

13:00–17:00	**Team Work Session 3 — Final Development** • Model optimisation and sensitivity analysis • Visualisation preparation (DAGs, result plots) • Drafting presentation slides and final report

	⚠ *Mentors available throughout the afternoon*

Day 3 — April 19: Finalisation, Presentations & Awards

09:00–12:00	**Team Work Session 4 — Finalisation** • Finalise code and ensure reproducibility • Complete final report • Polish presentation slides
12:00	**Submission Deadline** • Code repository • Final report • Presentation slides ⚠ *Submission format and platform to be confirmed*
12:00–13:00	**Lunch Break (self-arranged)**
12:00–15:30	**Judging Setup** • Judging takes place in the **East Wing** • On-site and remote evaluation **Judges:** Karim, Prof. Keivan Mallahi-Karai, Ahmed Maghri, Varun Patel *(online)*
15:30–17:00	**Team Presentations** • Format: 6 minutes presentation + Q&A per team • Each team presents: problem approach, causal methodology (DAGs / probabilistic models), and results ⚠ *Maha: confirm total number of teams to verify time allocation*
17:30–18:00	**Judging & Deliberation** • Innovation and creativity of causal approach • Technical quality (model correctness, DAG validity, probabilistic rigour) • Impact and real-world applicability of solution • Clarity and quality of presentation
18:00–18:20	**Awards Ceremony & Closing Remarks** • Announcement of winning teams • Special recognitions (if applicable) • Closing remarks by organisers

18:20–19:00	**Closing dinner & networking**

Sponsors & Partners

This Datathon is organised with the support of:

Host Institution	Industry Partner
Constructor University	**Insytes GmbH**

Logistics Notes (colab with Community Coordinator)

- Prepare and print badges, schedules, and team identifier materials
- Venue Day 1 (April 17): East Wing -- classroom-style layout (tables facing front, chairs arranged for talks and then team work)
- Venue Day 2 (April 18): IRC Seminar Room -- smaller space for support/mentoring, fewer attendees expected
- Venue Day 3 (April 19): East Wing -- same classroom-style layout for presentations and ceremony
- Table/seating: no rearrangement between days needed since different rooms are used for Day 2. East Wing layout stays the same on Day 1 and Day 3
- Equipment needed: 1 projector, microphones (for keynote, panel, presentations). East Wing has power outlets already -- bring a few extension cables as backup
- Decoration: none needed
- Setup: Math Society team handles room setup each morning
- Teardown Day 3: Math Society team handles after closing ceremony

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Funding and Operational Support

Funding: The Datathon is primarily funded by FAC .
Funding will cover:

- FAC covers: all food (snacks, drinks, lunch for externals via catering request), plus dinner and lunch via a separate request
- Insytes GmbH: no financial contribution -- contributing mentorship, challenge problem/dataset, and judging only
- Math Society: no direct funding needed -- awards produced for free via Makers Club
- No external sponsorship

Sponsorship: Currently, there is no external sponsorship confirmed for the event.

Insytes GmbH is not contributing financially; their contribution is entirely in-kind (mentorship, data, challenge problem, judging)

Operational Support: The following operational and technical support will be required:

What we need from the university:

- Microphones for talks, panel, and presentations
- Projector and screen
- Venue booking approval for East Wing (Days 1 and 3) and IRC Seminar Room (Day 2)
- Catering request processing (snacks/drinks via standard catering)

- Guidance on badge production process (TBC with Community Coordinator)

What Math Society handles:

- All room setup and teardown
- Badge design
- Award production (via Makers Club)
- Event hosting/MC duties
- Printing of schedules and materials
- Coordination with Insytes GmbH

Required Funding:

- Pizzas: A Pizza for 4 people costs 36 Euros, and for 40 students (the expected number), we need 10 pizzas. Therefore, the total price for Pizzas = 36*10=360 Euros.
- Other Food Expenses given in the table below.
- Catering request (via university): snacks, coffee, tea, small food items -- covered by FAC
- Lunch (Days 1 and 3): pizza or equivalent -- covered by FAC, separate request.
- Lunch (Day 2): self-arranged (lighter day, fewer attendees expected)
- Dinner: TBC -- needs separate request, covered by FAC.
- Awards: 3D printed trophies via Makers Club -- cost: 0 EUR
- Badges: designed by Math Society, printing/production process TBC (pending confirmation from Community Coordinator)
- Printing (schedules, materials): TBC -- check if university handles this
- Equipment: provided by university (microphones, projector) -- cost: 0 EUR

Items to be Purchased

Item Name	Number of Units	Price per Unit	Total	Categorey	Why this Item?
MadeGood Organic Chocolate Chip Granola Bars 6x24g	5	3.15	15.75	Food & Snacks	
Huober Organic Crispy Pretzel 175g	5	2.26	11.3	Food & Snacks	
REWE Organic Date and Peanut Bars 3x30g	4	1.79	7.16	Food & Snacks	
Tuc Cracker Paprika 100g	10	1.49	14.9	Food & Snacks	
Dr. Oetker Cinnamon Rolls Baking Mix 292g	5	3.14	15.7	Food & Snacks	
Yes! Embossed white napkins, 100 pieces	2	1.18	2.36	Disposables	
Yes! Embossed white napkins, 100 pieces	2	1.18	2.36	Disposables	
Gerolsteiner Medium 6x1.5l (ONE-WAY)	2	.94	1.88	Drinks	
Coca-Cola Zero Sugar 6x2l (ONE-WAY)	1	13.14	13.14	Drinks	
Volvic Juicy Orange-Mango 6x1l	1	11.14	11.14	Drinks	
Lorenz Crunchips Cheese & Onion 150g	10	1.99	19.9	Food & Snacks	
50x Hot and cold beverage cups, paper cups, mixed/PE white, 200ml, fill line Ø 70mm	2	5.19	10.38	Disposables	
PAPSTAR cardboard plates "pure" round 190 mm white 100 pieces	1	10.75	10.75	Disposables	

Total Amount from FAC 134.36€